Scalability & Future Plan

Date	6 July 2026
Team ID	
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Local JSON Caching	Cannot store large historical runs dynamically.	High
2	Manual Climate Inputs	Farmers have to lookup temperature/rainfall.	Medium
3	Web interface dependency	Difficult to access without active computer or phone.	Low

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load Capacity	Local Flask server handles single user.	Deploy using Gunicorn/Nginx on AWS EC2.
2	Database Scale	Static JSON file.	PostgreSQL database cluster.
3	Automation	Manual inputs.	IoT soil NPK/pH sensors streaming telemetry.
4	Security	Local server, no encryption or user login.	Implement JWT authentication, SSL/TLS, and secure parameter sanitization.

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 2	Weather APIs integration	Q3 2026	Auto-fetch rainfall/temp.
Phase 3	IoT Telemetry Hub	Q1 2027	Direct soil measurements.

Phase 4	Fertilizer Recommendation System	Q2 2027	Provides exact fertilizer bag count estimates based on soil deficiencies.